REFERENCE · SURVEY123 AND KOBOTOOLBOX

XLSForm cheat sheet

The question types, rules and expressions used most often when building forms for Survey123, KoboToolbox and ODK.

Question types

type	Use it for
`text`	Free text
`integer / decimal`	Whole or decimal numbers
`date / time / dateTime`	Date and time pickers
`select_one list`	One answer from a choices list
`select_multiple list`	Several answers from a list
`geopoint`	GPS location
`image / audio`	Photo or recording
`barcode`	Scan a barcode or QR code
`calculate`	Hidden value worked out by a formula
`note`	Text shown to the user, no answer
`begin group / end group`	Show questions together
`begin repeat / end repeat`	Ask a set of questions many times

Columns on the survey sheet

Column	Example	What it does
`required`	yes	Answer cannot be skipped
`relevant`	${sex} = 'female'	Show only when the condition is true
`constraint`	. >= 0 and . <= 120	The dot means "this answer"
`constraint_message`	Age must be 0 to 120	What the user sees when a rule fails
`calculation`	${doses_given} - ${doses_wasted}	Used with type calculate
`choice_filter`	district = ${district}	Cascading lists, such as district then chiefdom
`default`	today()	Pre-filled value
`appearance`	minimal · autocomplete · multiline	Changes how a question looks

Useful expressions

Expression	What it does
`selected(${symptoms}, 'fever')`	True if "fever" was ticked in a select_multiple
`count-selected(${symptoms})`	How many options were ticked
`regex(., '^[0-9]{9}$')`	Exactly nine digits, useful for phone numbers
`string-length(.) >= 3`	At least three characters
`. <= today()`	Date cannot be in the future
`int((today() - ${dob}) div 365.25)`	Approximate age in years from a date of birth
`if(${age} < 5, 'under5', 'over5')`	Choose a value based on a condition
`concat(${first}, ' ', ${last})`	Join text values

Required sheets

`survey`	One row per question: type, name, label and rules
`choices`	list_name, name, label for every select list
`settings`	form_title, form_id and version

Common mistakes

Check before you publish		
☐	Every name is unique, has no spaces and starts with a letter	Yes No N/A
☐	Every select list used on the survey sheet exists on the choices sheet	Yes No N/A
☐	Choice names are short codes, with the readable text in label	Yes No N/A
☐	Every relevant, constraint and calculation references ${names} that exist	Yes No N/A
☐	Phone numbers and IDs are text, not integer, so leading zeros are kept	Yes No N/A
☐	The version number was increased before republishing	Yes No N/A
☐	No question collects data you will not use, especially names and phone numbers	Yes No N/A

Tip. Test the form on a real phone, offline, with the person who will use it in the field, before training the whole team.